Parametric Inference

Every Exercise Covered in Class, with Answers

Compiled from lecture notes 1–10 (24 Jul – 28 Aug 2026)

This document collects *everything the professor set or worked in class*, lecture by lecture.

- **Part I** lists the worked examples with their results — what was done on the board, compressed to statement plus answer, so you can use it as a recall drill.
- **Part II** gives full solutions to the ■ exercises, i.e. every item the notes mark with a "#" and leave to you.

Companion documents: `PI_Exam_Revision.pdf` (theory and cheat sheets) and `PI_Assignment_Solutions.pdf` (Homeworks 1–3).

Discrepancy with the notes

Two places where the notes and the mathematics disagree. Both are flagged again in place.

(i) **Lecture 1, p.3.** The minimisation of $R(\sigma^2, T_c)$ lands on $c = -2/(n+1)$, which is struck through; p.4 has the correct $c = 1/(n+1)$. Use $1/(n+1)$.

(ii) **Lecture 7, "N.B. problem".** The note reads "No unbiased estimator exists for θ!" for the negative binomial $f(x \mid \theta) = (x-1)\theta^{x-2}(1-\theta)^2$. An unbiased estimator *does* exist, namely $(X-2)/(X-1)$; what has no unbiased estimator is $1/\theta$. Verified symbolically and numerically in §7 of Part II. Worth checking with the professor before the exam.

Contents

Part I — Worked examples, with results

Lecture 1 (24 Jul) — Risk, unbiasedness, UMVUE

Worked in class	**Result**
$X \sim \text{Bin}(n, \theta)$: can a uniformly "best" estimator exist?	No. If $R(\theta, T^*) \le R(\theta, T)$ for all T, θ, compare with $T \equiv \theta_0$ at $\theta = \theta_0$: $R(\theta_0, T^*) = 0 \Rightarrow T^* \equiv \theta_0$, which then fails at $\theta = \theta_0/2$.
Decomposition of $\mathbb{E}_\theta(T-\theta)^2$	$= \text{Bias}(T)^2 + \text{Var}_\theta(T)$.
Ex 1. $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bern}(\theta)$, $T^* = \overline{X}$	Unbiased; attains the CRLB; UMVUE with Var $= \theta(1-\theta)/n$.
Is the UMVUE unique?	Yes. If T_1, T_2 are both UMVUE with variance σ^2, then $T = (T_1+T_2)/2$ is unbiased and $\sigma^2 \le \text{Var}(T) = \frac{\sigma^2}{4}(2+2\rho)$ forces $\rho = 1$, i.e. $T_1 = T_2$.
Ex 2. $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\theta, 1)$, $T = \overline{X}$	$\text{Var}(\overline{X}) = 1/n = \text{CRLB} \Rightarrow \overline{X}$ is the unique UMVUE.
Ex 3. $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\theta, \sigma^2)$, both unknown	$S^2 = \frac{1}{n-1}\sum(X_i - \overline{X})^2$ is unbiased for σ^2; $\frac{1}{n}\sum(X_i - \overline{X})^2$ is biased but has smaller risk.
Ex 4. $X_1, \ldots, X_n \overset{\text{iid}}{\sim} U[0, \theta]$; two unbiased estimators $T_1 = 2\overline{X}$, $T_2 = \frac{n+1}{n} X_{(n)}$	$R(\theta, T_1) = \theta^2/(3n)$; $R(\theta, T_2)$ is Exercise 1.4 below.

Lecture 2 (29 Jul) — Estimability, sufficiency, completeness

Worked in class	**Result**
Ex 1. $X \sim \text{Bin}(n, \theta)$, estimate $\psi(\theta) = \frac{1-\theta}{\theta}$	No unbiased estimator: $\mathbb{E}_\theta T$ is a polynomial in θ of degree $\le n$ and cannot equal $1/\theta - 1$ on $(0, 1)$.
Ex 2. $X \sim \text{Poi}(\theta)$: which ψ are estimable?	Exactly the analytic ψ; then T is unique, hence the UMVUE.
Ex 3. $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \frac{1}{\theta} e^{-x/\theta}$	$\overline{X}$ unbiased, $\text{Var}(\overline{X}) = \theta^2/n = \text{CRLB} \Rightarrow$ unique UMVUE.
Is $T(X) = X$ the unique unbiased estimator when $n = 1$?	Yes: $\int_0^\infty T(x) e^{-x/\theta} dx = \theta^2 \ \forall\theta$ determines T by uniqueness of the Laplace transform.
Definition of sufficiency	Conditional law of $\mathbf{X}$ given T is free of θ.
Ex 4. $\text{Bern}(\theta)$, $T = \sum X_i$	$\mathbb{P}(\mathbf{X} = \mathbf{x} \mid T = t) = 1/\binom{n}{t}$ — Exercise 2.3.
Ex 5. $\text{Poi}(\theta)$, $T = \sum X_i$	$\text{Multinomial}(t; \frac{1}{n}, \ldots, \frac{1}{n})$ — Exercise 2.4.
Ex 6. $\text{Exp}(\theta)$, $T = \sum X_i$	Sufficient — Exercise 2.5. Completeness follows from Laplace-transform uniqueness (Exercise 2.6(c)).
Definition of completeness	$\mathbb{E}_\theta g(T) = 0 \ \forall\theta \Rightarrow g \equiv 0$; gives uniqueness of unbiased $h(T)$.

Lecture 3 (31 Jul) — Rao–Blackwell, Lehmann–Scheffé, factorisation

Worked in class	**Result**
Rao–Blackwell theorem	$T^* = \mathbb{E}[T \mid S]$ has the same bias and $\mathrm{Var}(T) = \mathbb{E}[\mathrm{Var}(T \mid S)] + \mathrm{Var}(T^*) \geq \mathrm{Var}(T^*)$. Strict unless T is already a function of S.
Lehmann–Scheffé theorem	Proof is Exercise 3.1.
Ex 1. Bern(θ)	$\overline{X}$ is THE UMVUE.
Ex 2. Poi(θ)	$S = \sum X_i \sim \mathrm{Poi}(n\theta)$; S/n is THE UMVUE.
Ex 3. $U[0, \theta]$	$S = X_{(n)}$; $\frac{n+1}{n}S$ is THE UMVUE.
Ex 4. Exp(θ)	$S = \sum X_i \sim \mathrm{Gamma}(n, \theta)$; S/n is THE UMVUE.
Neyman–Fisher factorisation	$f(\mathbf{x} \mid \theta) = h(\mathbf{x})g(\theta, S(\mathbf{x})) \iff S$ sufficient.
Ex 3 revisited. $U[0, \theta]$ via NFFT	$\theta^{-n}\mathbb{I}[x_{(1)} > 0]\,\mathbb{I}[x_{(n)} < \theta]$, so $X_{(n)}$ is sufficient.
Ex 5. $N(\theta, 1)$ via NFFT	$h = e^{-\frac{1}{2}\sum x_i^2}/(2\pi)^{n/2}$, $g = \exp\{\theta \sum x_i - \frac{n}{2}\theta^2\}$; $S = \sum X_i \sim N(n\theta, n)$, S/n is THE UMVUE.
Ex 6. Beta(θ, ϕ)	$S = \left(\prod X_i,\ \prod(1 - X_i)\right)$ is sufficient.
Ex 7. Cauchy(θ)	$S =$ the order statistic is sufficient.
General note	The order statistic is *always* sufficient for iid data with a pdf.

Lecture 4 (4 Aug) — Ancillarity, exponential families, Basu

Worked in class	**Result**
Ex 1. Cauchy(θ): is the order statistic complete?	No. $g(X_{(i)} - X_{(j)})$ has a θ-free distribution, so $\mathbb{E}_\theta[g(X_{(i)} - X_{(j)}) - c] \equiv 0$ with $g - c \not\equiv 0$. It is *ancillary*.
Definitions	Ancillary statistic; zero function.
Exponential family	(i) support free of θ; (ii) $f = p(\theta)e^{c(\theta)T(x)}h(x)$.
Ex 2. Bin(n, θ)	$p = (1-\theta)^n$, $c = \log\frac{\theta}{1-\theta}$, $T = x$, $h = \binom{n}{x}$.
Ex 3. Poi(θ)	$p = e^{-\theta}$, $c = \log\theta$, $T = x$, $h = 1/x!$.
Ex 4. $N(\theta, 1)$	$p = e^{-\theta^2/2}/\sqrt{2\pi}$, $c = \theta$, $T = x$, $h = e^{-x^2/2}$.
Ex 5. Cauchy(θ)	Not an exponential family — Exercise 4.1.
Ex 6. Gamma(ν known$, \theta$)	$p = (\Gamma(\nu)\theta^\nu)^{-1}$, $c = -1/\theta$, $T = x$, $h = x^{\nu-1}$.
Completeness criterion	If the natural parameter space $\Gamma = \{c(\theta)\}$ contains an interval with non-empty interior, $S = \sum T(X_i)$ is complete.
Power series family $a_x\theta^x/A(\theta)$	Exponential family with $c = \log\theta$, $\Gamma = (-\infty, \log b) \Rightarrow \sum X_i$ complete.
$N(\theta, \sigma^2)$, σ^2 known	$\overline{X}$ complete sufficient, S^2 ancillary; the hint chain $\mathbb{P}[S \leq s \mid T] = g(T)$, $\mathbb{E}_\theta g(T) = \mathbb{P}[S \leq s]$, completeness $\Rightarrow g \equiv \mathbb{P}[S \leq s]$ proves independence.
Basu's theorem	A complete sufficient statistic is independent of every ancillary statistic.

Consistency	$T_n \xrightarrow{p} \psi(\theta)$; UMVUE consistency is Exercise 4.2.

Lecture 5 (7 Aug) — Multiparameter families; into testing

Worked in class	**Result**
Ex 1. $N(\mu, \sigma^2)$	$T = (\sum X_i, \sum X_i^2)$ sufficient; natural space $\mathbb{R} \times (-\infty, 0)$ contains a rectangle $\Rightarrow$ complete.
Multiparameter exponential family	$S = \left(\sum T_1(X_i), \ldots, \sum T_k(X_i)\right)$; complete if the natural space contains a k-dimensional rectangle with nonempty interior.
Ex 2. Multinomial	Is in the family, with $c_i(\theta) = \log \frac{p_i}{1-p_1-\cdots-p_{k-1}}$, $T_i = Y_i$, $p(\theta) = (1 - p_1 - \cdots - p_{k-1})^n$, $h = n!/\prod y_i!$; $S = (T_1, \ldots, T_{k-1})$ complete sufficient.
Ex 3. $U[\phi, \psi]$	$(X_{(1)}, X_{(n)})$ sufficient; joint density $n(n-1)(v-u)^{n-2}/(\psi - \phi)^n$ — Exercise 5.1 — and complete.
Ex 4. $U[\theta, \theta + 1]$	$(X_{(1)}, X_{(n)})$ sufficient but *not* complete (Exercise 5.2); $\overline{X} - \frac{1}{2}$ is unbiased and Rao–Blackwellises to $\frac{1}{2}(X_{(1)} + X_{(n)} - 1)$ (Exercise 5.3); not the UMVUE, since the statistic is incomplete.
Ex 5. $N(\theta, 1)$, $\Theta = \{\theta_0, \theta_1\}$	Test function $\phi : \mathcal{X} \to [0, 1]$; power $= \mathbb{E}_{\theta_1}\phi$; size $= \mathbb{E}_{\theta_0}\phi =$ Type I error; Type II error $= 1-$ power.
Two-point parameter space	$T(x) = f_{\theta_1}(x)/f_{\theta_0}(x)$ is sufficient (take $h = f_{\theta_0}$).

Lectures 6–7 (11–14 Aug) — Testing setup; Bayes

Worked in class	**Result**
Sufficiency reduction in testing	$\psi(T) = \mathbb{E}[\phi \mid T]$ has the same size and power as ϕ, so only tests that are functions of a sufficient statistic need be considered.
Simple vs simple	Reject when $f(x \mid \theta_1)/f(x \mid \theta_0) > 1$; most powerful of its size — Exercise 6.1.
Bayes estimate	The posterior mean $\mathbb{E}[\theta \mid X]$.
Can a Bayes estimate be unbiased?	No except degenerately: conditioning two ways gives $\mathbb{E}_m[T\theta] = \mathbb{E}[\theta^2] = \mathbb{E}[T^2]$, so $\mathbb{E}_m[(T - \theta)^2] = 0$.
Posterior and sufficiency	$h(\mathbf{x})$ cancels, so the posterior — and every (generalized) Bayes estimate — is a function of the sufficient statistic.
Bayes risk	$r(\pi, T) = \int R(\theta, T)\pi(\theta)d\theta$, minimised at the posterior mean; minimum value $= \mathbb{E}_m[\mathrm{Var}(\theta \mid X)]$.
Coin toss $HHHTHHT$	$\hat{\theta} = 5/7$.
Improper prior $\pi \equiv 1$, $X \sim N(\theta, 1)$	Posterior $N(x, 1)$; generalized Bayes estimate X, and $\overline{X}$ for n iid observations.

Cauchy ML equation	$\sum_i \frac{2(x_i-\theta)}{1+(x_i-\theta)^2} = 0$: a polynomial equation of degree $2n-1$ with no closed-form solution.
Cauchy generation	$X = \tan\{\pi(U - \frac{1}{2})\} + \theta$ with $U \sim U[0,1]$.
Unbiased estimate for Cauchy, $n = 1$	None exists — discussed in Exercise 7.3.

Lectures 8–9 (21–25 Aug) — MLE machinery, minimax, NP, MLR

Worked in class	**Result**
Method of scoring	Solve $\sum_i f'(X_i \mid \theta)/f(X_i \mid \theta) = 0$ by Newton–Raphson, replacing the second derivative by its expectation, the Fisher information.
MLE and sufficiency	By NFFT the likelihood is $h(\mathbf{x})g(\theta, T)$, so the MLE is a function of T.
Invariance	$\psi(\hat{\theta})$ is unbiased when ψ is linear (same for Bayes); for the MLE, ψ may be any one-one function.
Minimax definition	$\sup_\theta R(\theta, T^*) = \inf_T \sup_\theta R(\theta, T)$.
Ex 1. $\text{Bin}(n, \theta)$, $\text{Beta}(p, q)$ prior	Bayes estimate $\frac{p+X}{p+q+n}$ (Exercise 8.2); UMVUE X/n has $\max_\theta R = \frac{1}{4n}$ (Exercise 8.3).
Constant-risk criterion	Bayes + constant risk $\Rightarrow$ minimax — Exercise 8.4.
Minimax for the Binomial	$p = q = \sqrt{n}/2$ gives $T^* = \frac{X+\sqrt{n}/2}{n+\sqrt{n}}$, constant risk $\frac{1}{4(1+\sqrt{n})^2}$, hence minimax.
Limit of Bayes rules	$r(\pi_k, T_{\pi_k}) \to c$ and $\sup_\theta R(\theta, T^*) = c \Rightarrow T^*$ minimax.
$U[a, \theta]$, $\theta \geq a$	$X_{(n)}$ sufficient with density nx^{n-1}/θ^n; Pareto $\pi(\theta) = c(\alpha, a)\theta^{-\alpha}$ is conjugate. Every estimator has infinite maximum risk — Exercise 9.1.
Neyman–Pearson lemma	$\phi = 1, \gamma, 0$ as $f_1/f_0 >, =, < k$; most powerful of its size.
MP test for $N(\theta, 1)$	$\phi = 1$ iff $X > c_\alpha$ with $\mathbb{P}_{\theta_0}(X > c_\alpha) = \alpha$; c_α does not involve θ_1.
MLR and $\text{Bin}(n, \theta)$	$\frac{\theta_1^x(1-\theta_1)^{n-x}}{\theta_0^x(1-\theta_0)^{n-x}}$ is increasing in x: MLR in x.
Exponential family $\subseteq$ MLR	$p(\theta)e^{c(\theta)T(x)}h(x)$ has MLR in $T(x)$ whenever c is monotone in θ.
Cauchy $\notin$ MLR	The ratio of two quadratics tends to 1 as $x \to \pm\infty$, so it cannot be monotone.

Lecture 10 (28 Aug) — Consistency

Worked in class	**Result**
b_n-consistency	$b_n(T_n - \theta) = O_p(1)$, i.e. $T_n - \theta = O_p(1/b_n)$.

Sample median	With $F(\theta) = \frac{1}{2}$, F strictly increasing and continuous near θ: $\mathbb{P}(\tilde{X}_n - \theta > \varepsilon) = \mathbb{P}\{\mathrm{Bin}(n, 1 - F(\theta + \varepsilon)) \geq n/2\} \to 0$ since $1 - F(\theta + \varepsilon) < \frac{1}{2}$, and symmetrically. So $\tilde{X}_n \xrightarrow{p} \theta$.
Cauchy remark	Always start from the sample median.
Simple linear regression	$\hat{\beta}_{LS} = \frac{\sum(x_i - \bar{x})Y_i}{\sum(x_i - \bar{x})^2}$ is unbiased with variance $\sigma^2 / \sum(x_i - \bar{x})^2$: consistent iff $\sum(x_i - \bar{x})^2 \to \infty$. $\hat{\alpha}_{LS}$ is Exercise 10.1.
$U(0, \theta)$	$\hat{\theta}_n = X_{(n)}$, $\mathbb{E}X_{(n)} = \frac{n\theta}{n+1}$; $\mathbb{P}(\hat{\theta}_n > \theta + \varepsilon) = 0$ and $\mathbb{P}(\hat{\theta}_n < \theta - \varepsilon) = \left(1 - \frac{\varepsilon}{\theta}\right)^n \to 0$, so consistent.
MLE consistency in general	Exercise 10.2.

Part II — The ■ exercises, solved

1 Lecture 1

■ **Exercise 1.1.** *In Ex 1, show $T^* = \overline{X}$ attains the Cramér–Rao lower bound, hence is the UMVUE, with variance $\theta(1-\theta)/n$.*

Answer. $\overline{X}$ is unbiased and $\operatorname{Var}_\theta(\overline{X}) = \theta(1-\theta)/n$. For the bound, with $f(x \mid \theta) = \theta^x(1-\theta)^{1-x}$ on $\{0,1\}$,

$$\log f = x\log\theta + (1-x)\log(1-\theta), \qquad \frac{\partial}{\partial\theta}\log f = \frac{x}{\theta} - \frac{1-x}{1-\theta} = \frac{x-\theta}{\theta(1-\theta)}.$$

Hence $I(\theta) = \operatorname{Var}_\theta\big(\frac{X-\theta}{\theta(1-\theta)}\big) = \dfrac{\theta(1-\theta)}{\theta^2(1-\theta)^2} = \dfrac{1}{\theta(1-\theta)}$, and

$$\text{CRLB} = \frac{1}{n\,I(\theta)} = \frac{\theta(1-\theta)}{n} = \operatorname{Var}_\theta(\overline{X}).$$

Equality in the CRLB means no unbiased estimator can do better at any θ, so $\overline{X}$ is the UMVUE. (Consistently, the score has the form $k(\theta)[T-\psi(\theta)]$ with $k = n/(\theta(1-\theta))$ and $T = \overline{X}$, which is exactly the attainment condition.)

■ **Exercise 1.2.** *Derive $\mathbb{E}_\theta[(T-\theta)^2] = [\mathbb{E}_\theta T - \theta]^2 + \operatorname{Var}_\theta(T)$.*

Answer. Insert and subtract $\mathbb{E}_\theta T$:

$$\mathbb{E}_\theta(T-\theta)^2 = \mathbb{E}_\theta\big[(T-\mathbb{E}_\theta T) + (\mathbb{E}_\theta T - \theta)\big]^2 = \operatorname{Var}_\theta(T) + (\mathbb{E}_\theta T-\theta)^2 + 2(\mathbb{E}_\theta T - \theta)\,\mathbb{E}_\theta[T - \mathbb{E}_\theta T],$$

and the last expectation is 0. The cross term vanishing is the only content.

■ **Exercise 1.3.** *For $T_c = c\sum_{i=1}^n (X_i - \overline{X})^2$ with $X_i \overset{iid}{\sim} N(\mu,\sigma^2)$, find the c minimising the risk.*

Answer. Write $W = \sum(X_i - \overline{X})^2$, so $W/\sigma^2 \sim \chi^2_{n-1}$ and

$$\mathbb{E}W = (n-1)\sigma^2, \qquad \operatorname{Var}W = 2(n-1)\sigma^4, \qquad \mathbb{E}W^2 = \operatorname{Var}W + (\mathbb{E}W)^2 = \sigma^4\big[2(n-1)+(n-1)^2\big] = \sigma^4(n^2-1).$$

Therefore

$$R(\sigma^2, T_c) = \mathbb{E}(cW-\sigma^2)^2 = c^2\mathbb{E}W^2 - 2c\sigma^2\mathbb{E}W + \sigma^4 = \sigma^4\big[c^2(n^2-1) - 2c(n-1) + 1\big].$$

Differentiating in c: $2c(n^2-1) - 2(n-1) = 0$, so

$$\boxed{c^* = \frac{n-1}{n^2-1} = \frac{1}{n+1}}, \qquad R_{\min} = \sigma^4\Big[1 - \frac{n-1}{n+1}\Big] = \frac{2\sigma^4}{n+1}.$$

Compare with the unbiased choice $c = \frac{1}{n-1}$, whose risk is $\frac{2\sigma^4}{n-1}$: strictly larger for every $n \geq 2$ and every σ^2. **The UMVUE of σ^2 is inadmissible.**

Discrepancy with the notes

The notes' first pass gives $c = -2/(n+1)$ (struck through) — a sign slip in $2c(n^2-1) - 2(n-1) = 0$. Page 4 has the correct $c = 1/(n+1)$.

■ **Exercise 1.4.** *Compute $R(\theta, T_2)$ for $T_2 = \frac{n+1}{n}X_{(n)}$ with $X_1,\ldots,X_n \overset{iid}{\sim} U[0,\theta]$.*

Answer. $X_{(n)}$ has cdf $(x/\theta)^n$ and density nx^{n-1}/θ^n on $[0,\theta]$, so

$$\mathbb{E}X_{(n)} = \frac{n\theta}{n+1}, \qquad \mathbb{E}X_{(n)}^2 = \int_0^\theta x^2\frac{nx^{n-1}}{\theta^n}dx = \frac{n\theta^2}{n+2},$$

$$\operatorname{Var} X_{(n)} = \frac{n\theta^2}{n+2} - \frac{n^2\theta^2}{(n+1)^2} = \frac{n\theta^2}{(n+1)^2(n+2)}.$$

T_2 is unbiased, so its risk is its variance:

$$\boxed{R(\theta, T_2) = \left(\frac{n+1}{n}\right)^2 \frac{n\theta^2}{(n+1)^2(n+2)} = \frac{\theta^2}{n(n+2)}}$$

against $R(\theta, T_1) = \theta^2/(3n)$ for $T_1 = 2\overline{X}$. The ratio is $R(T_1)/R(T_2) = (n+2)/3$, so T_2 wins for every $n \geq 2$ and by a factor growing linearly in n: the extreme order statistic extracts far more information than the mean here. (Simulation at $\theta = 3$: $n = 10$ gives empirical risk 0.0751 against the formula's 0.0750.)

2 Lecture 2

■ **Exercise 2.1.** $X \sim \mathrm{Bin}(n, \theta)$. *Show $\psi(\theta)$ has an unbiased estimator* iff *ψ is a polynomial of degree $\leq n$.*

Answer. ($\Rightarrow$) If $\mathbb{E}_\theta T = \psi(\theta)$ then $\psi(\theta) = \sum_{i=0}^n T(i)\binom{n}{i}\theta^i(1-\theta)^{n-i}$, a finite linear combination of polynomials of degree $\leq n$, hence itself such a polynomial.

($\Leftarrow$) Suppose $\psi(\theta) = \sum_{k=0}^n a_k\theta^k$. The falling factorial $X^{\underline{k}} = X(X-1)\cdots(X-k+1)$ satisfies

$$\mathbb{E}_\theta\left[X^{\underline{k}}\right] = n^{\underline{k}}\,\theta^k \qquad (0 \leq k \leq n),$$

so $X^{\underline{k}}/n^{\underline{k}}$ is unbiased for θ^k. Then $T(X) = \sum_{k=0}^n a_k X^{\underline{k}}/n^{\underline{k}}$ is unbiased for ψ. □

This immediately settles Ex 1: $\psi(\theta) = 1/\theta - 1$ is not a polynomial on $(0,1)$, so no unbiased estimator exists.

■ **Exercise 2.2.** $X \sim \mathrm{Poi}(\theta)$. *Check that ψ is unbiasedly estimable iff it is analytic.*

Answer. $\mathbb{E}_\theta T = \psi(\theta)$ for all $\theta > 0$ is equivalent to

$$\sum_{i\geq 0} \frac{T(i)}{i!}\,\theta^i = \psi(\theta)\,e^\theta, \qquad \theta > 0.$$

The left side is a power series in θ. So a solution T exists iff $\psi(\theta)e^\theta$ admits a power series expansion valid on $(0, \infty)$, i.e. iff ψe^θ is analytic; and since e^θ is analytic and never zero, this holds iff ψ is analytic. When it does, the coefficients of a power series are unique, so

$$T(i) = i! \cdot \left[\theta^i\right]\left(\psi(\theta)e^\theta\right)$$

is the *unique* unbiased estimator — and therefore the UMVUE, provided its variance is finite.

■ **Exercise 2.3.** $X_1, \ldots, X_n \overset{iid}{\sim} \mathrm{Bern}(\theta)$, $T = \sum X_i$. How *is the conditional distribution obtained?*

Answer. For $\mathbf{x} \in \{0,1\}^n$ with $\sum x_i = t$,

$$\mathbb{P}_\theta(\mathbf{X} = \mathbf{x} \mid T = t) = \frac{\mathbb{P}_\theta(\mathbf{X} = \mathbf{x})}{\mathbb{P}_\theta(T = t)} = \frac{\theta^t(1-\theta)^{n-t}}{\binom{n}{t}\theta^t(1-\theta)^{n-t}} = \frac{1}{\binom{n}{t}},$$

and 0 if $\sum x_i \neq t$. The θ's cancel exactly because every $\mathbf{x}$ with the same total has the same probability. Free of $\theta \Rightarrow T$ sufficient.

■ **Exercise 2.4.** $X_1, \ldots, X_n \overset{iid}{\sim} \mathrm{Poi}(\theta)$, $T = \sum X_i$. How*?*

Answer. $T \sim \mathrm{Poi}(n\theta)$, so for $\mathbf{x}$ with $\sum x_i = t$,

$$\mathbb{P}_\theta(\mathbf{X} = \mathbf{x} \mid T = t) = \frac{\prod_i e^{-\theta}\theta^{x_i}/x_i!}{e^{-n\theta}(n\theta)^t/t!} = \frac{e^{-n\theta}\theta^t/\prod_i x_i!}{e^{-n\theta}n^t\theta^t/t!} = \frac{t!}{x_1!\cdots x_n!}\left(\frac{1}{n}\right)^t.$$

This is the Multinomial$\left(t; \frac{1}{n}, \dots, \frac{1}{n}\right)$ pmf — free of θ, so T is sufficient.

■ **Exercise 2.5.** $X_1, \dots, X_n \overset{iid}{\sim} \frac{1}{\theta} e^{-x/\theta}$, $T = \sum X_i$. *Show T is sufficient.* (Hint: do $n = 2$ first; the conditional distribution will be Beta.)

Answer. *Case $n = 2$ (the hint).* $T = X_1 + X_2 \sim \text{Gamma}(2, \theta)$ with density $f_T(t) = te^{-t/\theta}/\theta^2$. The joint density of (X_1, T) is $f(x_1, t - x_1) = \theta^{-2} e^{-t/\theta}$ on $0 < x_1 < t$. Hence

$$f_{X_1|T}(x_1 \mid t) = \frac{\theta^{-2} e^{-t/\theta}}{t\,\theta^{-2} e^{-t/\theta}} = \frac{1}{t}, \qquad 0 < x_1 < t,$$

so $X_1 \mid T = t \sim U(0, t)$, i.e. $X_1/T \sim U(0, 1) = \text{Beta}(1, 1)$ — free of θ, as the hint says.

General n. The joint density is

$$f(\mathbf{x} \mid \theta) = \theta^{-n} \exp\Big\{ -\frac{1}{\theta} \sum_i x_i \Big\} \prod_i \mathbb{I}[x_i > 0] = \underbrace{\theta^{-n} e^{-T(\mathbf{x})/\theta}}_{g(\theta,T)} \cdot \underbrace{\prod_i \mathbb{I}[x_i > 0]}_{h(\mathbf{x})},$$

so by the Neyman–Fisher factorisation theorem $T = \sum X_i$ is sufficient. (Directly: $(X_1, \dots, X_{n-1})/T$ given $T = t$ is uniform on the simplex, i.e. Dirichlet$(1, \dots, 1)$, so marginally $X_1/T \sim \text{Beta}(1, n - 1)$ — again free of θ.)

■ **Exercise 2.6.** *Are the sufficient statistics of Ex 4, Ex 5 complete?* (Use the polynomial argument.) *And what about Ex 6?*

Answer. (a) Bernoulli. $T \sim \text{Bin}(n, \theta)$. Suppose $\mathbb{E}_\theta g(T) = 0$ for all $\theta \in (0, 1)$:

$$\sum_{t=0}^{n} g(t) \binom{n}{t} \theta^t (1 - \theta)^{n-t} = 0.$$

Divide by $(1 - \theta)^n > 0$ and substitute $r = \theta/(1 - \theta)$, which ranges over $(0, \infty)$:

$$\sum_{t=0}^{n} g(t) \binom{n}{t} r^t = 0 \qquad \forall r > 0.$$

A polynomial in r vanishing on an interval is identically zero, so all coefficients vanish: $g(t)\binom{n}{t} = 0$, i.e. $g \equiv 0$. **Complete.**

(b) Poisson. $T \sim \text{Poi}(n\theta)$. If $\mathbb{E}_\theta g(T) = 0$ for all $\theta > 0$,

$$e^{-n\theta} \sum_{t \geq 0} g(t) \frac{(n\theta)^t}{t!} = 0 \implies \sum_{t \geq 0} \frac{g(t) n^t}{t!} \theta^t = 0 \qquad \forall \theta > 0.$$

A power series vanishing on an interval has all coefficients zero, so $g \equiv 0$. **Complete.**

(c) Exponential. $T \sim \text{Gamma}(n, \theta)$. If $\mathbb{E}_\theta g(T) = 0$ for all $\theta > 0$,

$$\int_0^\infty g(t)\, t^{n-1} e^{-t/\theta}\, dt = 0 \qquad \forall \theta > 0.$$

With $s = 1/\theta$ this says the Laplace transform of $t \mapsto g(t)t^{n-1}$ vanishes on $(0, \infty)$. By uniqueness of the Laplace transform, $g(t)t^{n-1} = 0$ for a.e. $t > 0$, hence $g = 0$ a.e. **Complete.** (This is the "we can't prove it yet" step in the notes; it is the same transform-uniqueness fact used for Ex 3.)

3 Lecture 3

■ **Exercise 3.1.** *Prove the Lehmann–Scheffé theorem.*

Answer. *Claim.* If S is complete sufficient and $h(S)$ is unbiased for $\psi(\theta)$, then $h(S)$ is the UMVUE and is a.s. unique.

Let T be any unbiased estimator of $\psi(\theta)$ with finite variance. Put $T^* = \mathbb{E}[T \mid S]$; sufficiency makes this a statistic (free of θ). By Rao–Blackwell, T^* is unbiased and $\mathrm{Var}_\theta(T^*) \le \mathrm{Var}_\theta(T)$ for every θ. Both T^* and $h(S)$ are unbiased functions of S, so

$$\mathbb{E}_\theta\big[T^* - h(S)\big] = 0 \qquad \forall\theta,$$

and completeness of S forces $T^* = h(S)$ a.s. Therefore $\mathrm{Var}_\theta(h(S)) = \mathrm{Var}_\theta(T^*) \le \mathrm{Var}_\theta(T)$ for every unbiased T and every θ: $h(S)$ is the UMVUE.

Uniqueness. If $h_1(S), h_2(S)$ are both unbiased then $\mathbb{E}_\theta[h_1 - h_2] = 0$ for all θ, so $h_1 = h_2$ a.s. by completeness. □

■ **Exercise 3.2.** *Identify the sufficient statistic in Ex 1, Ex 2 and Ex 4 using the Neyman–Fisher factorisation.*

Answer. In each case group the θ-dependence.

$$\textbf{Ex 1 (Bern):}\quad \prod_i \theta^{x_i}(1-\theta)^{1-x_i} = \underbrace{(1-\theta)^n\Big(\frac{\theta}{1-\theta}\Big)^{\sum x_i}}_{g(\theta,S)} \cdot \underbrace{1}_{h}, \qquad S = \sum X_i.$$

$$\textbf{Ex 2 (Poi):}\quad \prod_i \frac{e^{-\theta}\theta^{x_i}}{x_i!} = \underbrace{e^{-n\theta}\theta^{\sum x_i}}_{g(\theta,S)} \cdot \underbrace{\frac{1}{\prod_i x_i!}}_{h}, \qquad S = \sum X_i.$$

$$\textbf{Ex 4 (Exp):}\quad \prod_i \frac{1}{\theta}e^{-x_i/\theta} = \underbrace{\theta^{-n}e^{-\sum x_i/\theta}}_{g(\theta,S)} \cdot \underbrace{\prod_i \mathbb{I}[x_i > 0]}_{h}, \qquad S = \sum X_i.$$

In all three, $S = \sum X_i$ — consistent with the direct conditional-distribution computations of Lecture 2.

4 Lecture 4

■ **Exercise 4.1.** *Check why* $\mathrm{Cauchy}(\theta)$ *is* not *in the exponential family.*

Answer. Two arguments; the second is the one to write down.

(a) Mixed-partial test. If $f(x \mid \theta) = p(\theta)e^{c(\theta)T(x)}h(x)$ then $\log f = \log p(\theta) + c(\theta)T(x) + \log h(x)$, so

$$\frac{\partial^2}{\partial\theta\,\partial x}\log f = c'(\theta)T'(x)$$

must *factorise* into a function of θ times a function of x. For the Cauchy, with $u = x - \theta$,

$$\log f = -\log\pi - \log(1+u^2), \qquad \frac{\partial}{\partial x}\log f = \frac{-2u}{1+u^2}, \qquad \frac{\partial^2}{\partial\theta\,\partial x}\log f = \frac{2(1-u^2)}{(1+u^2)^2}.$$

This vanishes exactly at $x = \theta \pm 1$ — zeros whose *location moves with* θ. A product $c'(\theta)T'(x)$ has its x-zeros at fixed positions independent of θ (or vanishes identically). Contradiction.

(b) Dimension of the sufficient statistic. In a one-parameter exponential family, $\sum_i T(X_i)$ is sufficient for every n: a *one-dimensional* sufficient statistic. For the Cauchy the minimal sufficient statistic is the full order statistic $(X_{(1)}, \dots, X_{(n)})$, whose dimension grows with n (Lecture 3, Ex 7). So the Cauchy cannot be an exponential family.

> **Note**
>
> Both failures matter later: because the Cauchy is not an exponential family it is also not an MLR family (Lecture 9), so no UMP test exists; and its sufficient statistic is not complete (Lecture 4, Ex 1), so Lehmann–Scheffé is unavailable.

■ **Exercise 4.2.** *Let T_n be the UMVUE of $\psi(\theta)$ based on $X_1, \dots, X_n$. Prove $T_n \xrightarrow{p} \psi(\theta)$.*

Answer. As stated the claim needs one hypothesis, which the argument makes clear: assume there exists *some* sequence $\tilde{T}_n$ of unbiased estimators of $\psi(\theta)$ with $\mathrm{Var}_\theta(\tilde{T}_n) \to 0$. (In every example in the course this is available — e.g. an average of iid unbiased estimators, whose variance is $O(1/n)$.)

Since T_n is the UMVUE and $\tilde{T}_n$ is unbiased,

$$0 \le \mathrm{Var}_\theta(T_n) \le \mathrm{Var}_\theta(\tilde{T}_n) \longrightarrow 0.$$

T_n is unbiased, so for any $\varepsilon > 0$ Chebyshev's inequality gives

$$\mathbb{P}_\theta\big(|T_n - \psi(\theta)| > \varepsilon\big) \le \frac{\mathrm{Var}_\theta(T_n)}{\varepsilon^2} \longrightarrow 0.$$

Hence $T_n \xrightarrow{p} \psi(\theta)$ for every θ. □

Without the hypothesis the statement can fail: if no unbiased sequence has vanishing variance, the UMVUE inherits nothing to be squeezed against.

5 Lecture 5

■ **Exercise 5.1.** $X_1, \dots, X_n \overset{iid}{\sim} U[\phi, \psi]$. How *do we get the joint distribution of* $(X_{(1)}, X_{(n)})$*, and why is it complete?*

Answer. Joint density. For $\phi < u < v < \psi$, the event $\{X_{(1)} \in du,\ X_{(n)} \in dv\}$ requires one observation in du, one in dv, and the remaining $n-2$ strictly between: choose which observation is the minimum (n ways) and which is the maximum ($n-1$ ways), so

$$f_{(1),(n)}(u,v) = n(n-1) f(u) f(v) \big[F(v) - F(u)\big]^{n-2} = n(n-1) \cdot \frac{1}{(\psi-\phi)^2} \cdot \frac{(v-u)^{n-2}}{(\psi-\phi)^{n-2}} = \frac{n(n-1)(v-u)^{n-2}}{(\psi-\phi)^n}.$$

Completeness. Suppose $\mathbb{E}_{\phi,\psi}\, g(X_{(1)}, X_{(n)}) = 0$ for all $\phi < \psi$:

$$\int_\phi^\psi \int_\phi^v g(u,v)\,(v-u)^{n-2}\,du\,dv = 0 \qquad \forall\, \phi < \psi$$

(the constant $n(n-1)/(\psi-\phi)^n$ is positive and drops out). Differentiate with respect to ψ (Leibniz, the inner integral evaluated at $v = \psi$):

$$\int_\phi^\psi g(u,\psi)(\psi-u)^{n-2}\,du = 0 \qquad \forall\, \phi < \psi.$$

Now differentiate with respect to ϕ:

$$-\,g(\phi,\psi)\,(\psi-\phi)^{n-2} = 0 \qquad \forall\, \phi < \psi.$$

Since $(\psi-\phi)^{n-2} > 0$, $g(\phi,\psi) = 0$ for all $\phi < \psi$. Hence $(X_{(1)}, X_{(n)})$ is **complete**.

■ **Exercise 5.2.** $X_1, \dots, X_n \overset{iid}{\sim} U[\theta, \theta+1]$. *Check that $X_{(n)} - X_{(1)}$ is free of θ (so the sufficient statistic is not complete).*

Answer. Write $X_i = \theta + U_i$ with $U_1, \dots, U_n \overset{\text{iid}}{\sim} U[0,1]$; this is an exact distributional identity because the family is a pure location family. Order statistics are monotone equivariant, so $X_{(k)} = \theta + U_{(k)}$ and

$$R := X_{(n)} - X_{(1)} = U_{(n)} - U_{(1)},$$

whose distribution involves no θ: R is **ancillary**, with density $f_R(r) = n(n-1)r^{n-2}(1-r)$ on $(0,1)$, i.e. $R \sim \mathrm{Beta}(n-1, 2)$, so $\mathbb{E}R = \frac{n-1}{n+1}$.

Consequently $g(u,v) := (v-u) - \frac{n-1}{n+1}$ satisfies $\mathbb{E}_\theta\, g(X_{(1)}, X_{(n)}) = 0$ for every θ, while $g \not\equiv 0$. The sufficient statistic $(X_{(1)}, X_{(n)})$ is therefore **not complete**, and Lehmann–Scheffé does not apply — which is exactly why the next exercise's estimator is not the UMVUE.

■ **Exercise 5.3.** *$\overline{X} - \frac{1}{2}$ is unbiased for θ.* How *does Rao–Blackwell give the better estimator $\frac{1}{2}\big(X_{(1)} + X_{(n)} - 1\big)$?*

Answer. We need $\mathbb{E}\big[\overline{X} \mid X_{(1)} = u,\ X_{(n)} = v\big]$.

Condition on the two extremes. Given $X_{(1)} = u$ and $X_{(n)} = v$, the remaining $n-2$ observations are distributed as $n-2$ iid $U(u,v)$ variables (in some order) — this is immediate from the joint density of all n order statistics, which is constant on $u < x_{(2)} < \cdots < x_{(n-1)} < v$. Each therefore has conditional mean $\frac{u+v}{2}$, so

$$\mathbb{E}\Big[\sum_{i=1}^{n} X_i \;\Big|\; u,v\Big] = u + v + (n-2)\cdot\frac{u+v}{2} = (u+v)\Big(1 + \frac{n-2}{2}\Big) = \frac{n(u+v)}{2},$$

and hence

$$\mathbb{E}\big[\overline{X} \mid u,v\big] = \frac{u+v}{2} \quad\Longrightarrow\quad \mathbb{E}\Big[\overline{X} - \tfrac{1}{2} \;\Big|\; u,v\Big] = \frac{X_{(1)} + X_{(n)} - 1}{2}.$$

By Rao–Blackwell this midrange-based estimator is unbiased with variance no larger than that of $\overline{X} - \frac{1}{2}$; the improvement is strict because $\overline{X}$ is not a function of $(X_{(1)}, X_{(n)})$. (Simulation, $n = 6$: $\mathrm{Var}(\overline{X}) = 0.0139$ versus $\mathrm{Var}\left(\frac{X_{(1)}+X_{(n)}}{2}\right) = 0.0089$.)

It is **not** the UMVUE: by Exercise 5.2 the conditioning statistic is not complete, so nothing forces uniqueness, and one can add any multiple of the zero-mean ancillary $g(X_{(1)}, X_{(n)})$ of Exercise 5.2 to obtain other unbiased estimators.

6 Lecture 6

■ **Exercise 6.1.** *Prove that the likelihood-ratio test has maximum power among all tests of the same size.* (Hint: let $\psi(x)$ indicate whether LR > 1; examine $(\psi(x) - \phi(x))(f_1(x) - f_0(x))$, find an inequality and integrate.)

Answer. This is the Neyman–Pearson lemma; the hint is exactly the proof. State it for a general cutoff $k \ge 0$ (the notes take $k = 1$).

Let ψ be the LR test, $\psi(x) = 1$ if $f_1(x) > kf_0(x)$ and 0 if $f_1(x) < kf_0(x)$, and let ϕ be any test with $\mathbb{E}_0\phi \le \mathbb{E}_0\psi$.

The pointwise inequality. Consider $\big(\psi(x) - \phi(x)\big)\big(f_1(x) - kf_0(x)\big)$.

- Where $f_1 > kf_0$: $\psi = 1 \ge \phi$, so the first factor is ≥ 0 and the second is > 0.
- Where $f_1 < kf_0$: $\psi = 0 \le \phi$, so the first factor is ≤ 0 and the second is < 0.
- Where $f_1 = kf_0$: the second factor is 0.

In every case the product is ≥ 0.

Integrate.

$$0 \;\le\; \int \big(\psi - \phi\big)\big(f_1 - kf_0\big) = \big(\mathbb{E}_1\psi - \mathbb{E}_1\phi\big) - k\big(\mathbb{E}_0\psi - \mathbb{E}_0\phi\big).$$

Since $k \ge 0$ and $\mathbb{E}_0\psi - \mathbb{E}_0\phi \ge 0$, the subtracted term is ≥ 0, so

$$\mathbb{E}_1\psi - \mathbb{E}_1\phi \;\ge\; k\big(\mathbb{E}_0\psi - \mathbb{E}_0\phi\big) \;\ge\; 0.$$

Thus $\mathbb{E}_1\psi \ge \mathbb{E}_1\phi$: the LR test is most powerful among all tests of its size or smaller. □

■ **Exercise 6.2.** *$X \sim \mathrm{Bin}(r,\theta)$ with the Beta family of priors $\pi(\theta) = \theta^{m-1}(1-\theta)^{n-1}/B(m,n)$. Calculate the posterior.*

Answer. Keep only the θ-dependence:

$$\pi(\theta \mid x) \;\propto\; \underbrace{\binom{r}{x}\theta^{x}(1-\theta)^{r-x}}_{\text{likelihood}} \cdot \underbrace{\frac{\theta^{m-1}(1-\theta)^{n-1}}{B(m,n)}}_{\text{prior}} \;\propto\; \theta^{m+x-1}(1-\theta)^{n+r-x-1}.$$

This is the kernel of a Beta density, so no integration is needed:

$$\boxed{\theta \mid X = x \;\sim\; \text{Beta}(m+x,\; n+r-x), \qquad \pi(\theta \mid x) = \frac{\theta^{m+x-1}(1-\theta)^{n+r-x-1}}{B(m+x,\; n+r-x)}}$$

with posterior mean — the Bayes estimate — equal to $\dfrac{m+x}{m+n+r}$.

■ **Exercise 6.3.** *$X \sim \text{Poi}(\theta)$: guess a conjugate family of priors.*

Answer. The likelihood in θ is $e^{-\theta}\theta^{x}$, so we want a prior family closed under multiplication by $e^{-\theta}\theta^{x}$: densities of the form $\theta^{\alpha-1}e^{-\theta/\lambda}$, i.e. the **Gamma family**

$$\pi(\theta) = \frac{\theta^{\alpha-1}e^{-\theta/\lambda}}{\Gamma(\alpha)\lambda^{\alpha}}, \qquad \alpha, \lambda > 0.$$

Then $\pi(\theta \mid x) \propto \theta^{\alpha+x-1}e^{-\theta(1+1/\lambda)}$, so

$$\theta \mid X = x \;\sim\; \text{Gamma}\Big(\alpha + x,\; \frac{\lambda}{1+\lambda}\Big), \qquad \text{Bayes estimate} = \frac{\lambda(\alpha+X)}{1+\lambda}.$$

For n iid observations replace x by $\sum x_i$ and 1 by n in the rate: $\text{Gamma}\big(\alpha + \sum x_i,\; \frac{\lambda}{1+n\lambda}\big)$.

■ **Exercise 6.4.** *$X \sim N(\theta, \sigma^2)$ with σ^2 known: what is a conjugate family?*

Answer. The **normal family** $\pi(\theta) = N(\mu, \tau^2)$. Both likelihood and prior are Gaussian kernels in θ, and the exponent is a sum of two quadratics:

$$-\frac{(x-\theta)^2}{2\sigma^2} - \frac{(\theta-\mu)^2}{2\tau^2} = -\frac{1}{2}\Big(\frac{1}{\sigma^2} + \frac{1}{\tau^2}\Big)\Big(\theta - \frac{\tau^2 x + \sigma^2\mu}{\sigma^2+\tau^2}\Big)^2 + \text{const.}$$

Hence

$$\boxed{\theta \mid X = x \;\sim\; N\Big(\frac{\tau^2 x + \sigma^2\mu}{\sigma^2+\tau^2},\; \frac{\sigma^2\tau^2}{\sigma^2+\tau^2}\Big)}$$

The posterior mean is a precision-weighted average of the data and the prior mean, and the posterior *precision* is the sum of the two precisions: $\frac{1}{\sigma^2} + \frac{1}{\tau^2}$. For n iid observations replace x by $\overline{X}$ and σ^2 by σ^2/n.

■ **Exercise 6.5.** *Identify a conjugate family for the Cauchy distribution.*

Answer. The Cauchy location family is not an exponential family (Exercise 4.1), so there is no finite-dimensional conjugate family of the usual kind. There *is* an honest conjugate family, obtained by closing the priors under the likelihood itself:

$$\boxed{\Pi = \Big\{\; \pi_{k,\mathbf{a}}(\theta) \;\propto\; \prod_{j=1}^{k} \frac{1}{1+(a_j-\theta)^2} \;:\; k \ge 1,\; a_1, \ldots, a_k \in \mathbb{R} \Big\}.}$$

Closure. With $X_1, \ldots, X_n$ observed,

$$\pi(\theta \mid \mathbf{x}) \;\propto\; \prod_{i=1}^{n} \frac{1}{1+(x_i-\theta)^2} \cdot \prod_{j=1}^{k} \frac{1}{1+(a_j-\theta)^2} = \prod_{\ell=1}^{k+n} \frac{1}{1+(b_\ell-\theta)^2},$$

where (b_ℓ) is the concatenation of (a_j) and (x_i). So the posterior lies in Π with k replaced by $k+n$: the family is genuinely conjugate.

Propriety. The integrand decays like $|\theta|^{-2k}$, so $\pi_{k,\mathbf{a}}$ is a proper density for every $k \ge 1$ ($k = 0$ gives the improper flat prior).

The catch, and why the professor kept pressing on this. Conjugacy here is bought by letting the number of hyperparameters grow with the sample size — the "prior" after n observations is described by $k+n$ numbers. This is the same objection raised in Lecture 7 against "take the class of all densities": a conjugate family is only useful when it is *small*. The Cauchy has no small one.

■ **Exercise 6.6.** *Is the Bayes estimate unbiased for θ in (1) Binomial with Beta prior, (2) Poisson with Gamma prior, (3) Normal with Normal prior?*

Answer. No in all three, and it is quickest to check directly.

(1) $T = \dfrac{m+X}{m+n+r}$ with $X \sim \mathrm{Bin}(r,\theta)$:

$$\mathbb{E}_\theta T = \frac{m+r\theta}{m+n+r},$$

which equals θ for all θ only if $r = m+n+r$ and $m=0$, i.e. $m=n=0$ — not a proper Beta prior.

(2) $T = \dfrac{\lambda(\alpha+X)}{1+\lambda}$ with $X \sim \mathrm{Poi}(\theta)$:

$$\mathbb{E}_\theta T = \frac{\lambda(\alpha+\theta)}{1+\lambda},$$

which equals θ for all θ only if $\frac{\lambda}{1+\lambda} = 1$ (impossible for finite λ).

(3) $T = \dfrac{\tau^2 X + \sigma^2\mu}{\sigma^2+\tau^2}$ with $X \sim N(\theta,\sigma^2)$:

$$\mathbb{E}_\theta T = \frac{\tau^2\theta + \sigma^2\mu}{\sigma^2+\tau^2},$$

which equals θ for all θ only if $\sigma^2 = 0$.

The general reason (the argument from the same lecture, worth memorising). Suppose T is both Bayes, $T = \mathbb{E}[\theta \mid X]$, and unbiased, $\mathbb{E}[T \mid \theta] = \theta$. Writing $\mathbb{E}_m$ for expectation under the joint law of (X,θ) and conditioning two different ways,

$$\mathbb{E}_m[T\theta] = \mathbb{E}\big[\theta\,\mathbb{E}(T \mid \theta)\big] = \mathbb{E}[\theta^2], \qquad \mathbb{E}_m[T\theta] = \mathbb{E}\big[T\,\mathbb{E}(\theta \mid X)\big] = \mathbb{E}[T^2].$$

Therefore

$$\mathbb{E}_m\big[(T-\theta)^2\big] = \mathbb{E}[T^2] + \mathbb{E}[\theta^2] - 2\mathbb{E}_m[T\theta] = 0,$$

forcing $T = \theta$ almost surely — impossible unless the problem is degenerate (the prior is a point mass, or θ is a function of X). In every non-trivial Bayes problem the Bayes estimate shrinks toward the prior mean, and shrinkage is precisely bias.

7 Lecture 7

■ **Exercise 7.1.** *Find a conjugate family of priors for an arbitrary density $f(x \mid \theta)$.*

Answer. Two answers, and the contrast is the point of the exercise.

Trivial answer. The class of *all* prior densities on Θ is conjugate: the posterior is a density on Θ, hence in the class. Useless, because a conjugate family earns its keep by being a small parametric family in which the updating rule is an explicit map on the hyperparameters.

Useful general construction. Fix a base measure π_0 and take

$$\Pi = \left\{ \pi(\theta) \ \propto\ \pi_0(\theta)\prod_{j=1}^{k} f(y_j \mid \theta) \ :\ k \ge 0,\ y_1,\dots,y_k \in \mathcal{X} \right\},$$

the priors that look like "π_0 plus k imaginary observations". Observing $x_1,\dots,x_n$ appends them to the list, so Π is conjugate for *any* f. The Cauchy family of Exercise 6.5 is exactly this construction with $\pi_0 \equiv 1$.

For an exponential family, this construction collapses: $\prod_j f(y_j \mid \theta)$ depends on the y_j only through $\sum_j T(y_j)$ and k, so Π is a genuinely *two*-parameter family — which is why Beta–Binomial, Gamma–Poisson and Normal–Normal are finite-dimensional and the Cauchy is not.

■ **Exercise 7.2.** *"N.B. problem". Toss until 2 tails:* $f(x \mid \theta) = (x-1)\theta^{x-2}(1-\theta)^2$, $x = 2, 3, \ldots$ *Does an unbiased estimator of* θ *exist?*

Answer. Yes — and it is unique, hence the UMVUE. Set $\mathbb{E}_\theta T = \theta$:

$$\sum_{x\ge 2} T(x)(x-1)\theta^{x-2}(1-\theta)^2 = \theta \quad\iff\quad \sum_{x\ge 2} T(x)(x-1)\theta^{x-2} = \frac{\theta}{(1-\theta)^2} \qquad \forall\theta\in(0,1).$$

Now expand the right side. Since $\sum_{k\ge1} k\theta^{k-1} = (1-\theta)^{-2}$,

$$\frac{\theta}{(1-\theta)^2} = \sum_{k\ge1} k\,\theta^k.$$

On the left, substitute $j = x-2$: $\sum_{j\ge0} T(j+2)(j+1)\theta^j$. Matching coefficients of θ^j:

$$j=0:\ T(2)\cdot 1 = 0; \qquad j\ge1:\ T(j+2)(j+1) = j.$$

Hence $T(j+2) = \dfrac{j}{j+1}$, i.e. writing $x = j+2$,

$$\boxed{T(X) = \frac{X-2}{X-1}}$$

(which also gives $T(2) = 0$, consistent with the $j = 0$ equation). *Verification:*

$$\mathbb{E}_\theta T = \sum_{x\ge2} \frac{x-2}{x-1}(x-1)\theta^{x-2}(1-\theta)^2 = (1-\theta)^2 \sum_{j\ge0} j\,\theta^j = (1-\theta)^2\cdot\frac{\theta}{(1-\theta)^2} = \theta.\ \checkmark$$

T is bounded ($0 \le T < 1$), so its variance is finite; uniqueness of the power-series coefficients makes it the only unbiased estimator, hence the UMVUE.

What *does* fail: $1/\theta$. Ask for $\mathbb{E}_\theta T = 1/\theta$:

$$\sum_{x\ge2} T(x)(x-1)\theta^{x-1} = \frac{1}{(1-\theta)^2} = \sum_{k\ge0}(k+1)\theta^k.$$

The left side is a power series with *no constant term* (the lowest power is θ^1), while the right side has constant term 1. No choice of T can match, so $1/\theta$ has no unbiased estimator. The same obstruction kills any ψ whose expansion has a non-zero constant term.

Discrepancy with the notes

The lecture note records "No Unbiased estimator exists for θ!" for this problem. As the computation and the check above show, that is not right for θ itself — $(X-2)/(X-1)$ works, and a numerical check at $\theta = 0.2, 0.5, 0.8$ reproduces θ to six decimals. The non-existence statement is correct for $1/\theta$. Confirm which one the professor intended before relying on it in an answer.

■ **Exercise 7.3.** $X_1,\ldots,X_n \overset{iid}{\sim} \text{Cauchy}(\theta)$. *Is there any unbiased estimate of* θ*? For* $n = 1$*?*

Answer. For $n = 1$: no. The obvious candidate fails immediately — X has no mean, since $\int |x|/\{\pi(1+(x-\theta)^2)\}dx = \infty$ — so X is not an unbiased estimator of anything. More is true: no T works at all. Unbiasedness says

$$g(\theta) := \int_{\mathbb{R}} T(x)\,\frac{1}{\pi\{1+(x-\theta)^2\}}\,dx = \theta \qquad \forall\theta\in\mathbb{R},$$

i.e. the convolution $T * f = \mathrm{id}$ where f is the standard Cauchy density. In Fourier terms $\widehat{f}(t) = e^{-|t|}$, so one would need $\widehat{T}(t) = e^{|t|} \cdot \widehat{(\cdot)}(t)$, which grows too fast at infinity to be the transform of any T for which the integral above converges absolutely for every θ. So no unbiased estimator exists.

Sketch you can write in an exam. Absolute convergence for all θ forces $\int |T(x)|/(1+x^2)\,dx < \infty$. Smoothing such a T by a Cauchy kernel produces a function that is bounded on compacta and cannot grow linearly without bound in θ; but $g(\theta) = \theta$ does. The full argument is the Fourier one above and is outside the course's toolkit — the notes assert it ("for $n = 1$, certainly not!") without proof, so a short version like this is what is expected.

For $n \geq 2$ unbiased estimators of θ do exist — for instance a trimmed mean, or the sample median for odd n, both of which have finite expectation equal to θ by symmetry. What is not available is a UMVUE, because the sufficient order statistic is not complete (Exercise 4.1 and Lecture 4, Ex 1).

8 Lecture 8

■ **Exercise 8.1.** *Solve Q3 of PS1 when both θ and σ^2 are unknown.*

Answer. Homework 1 Q3 asks for the UMVUE of $\mathbb{P}\{X_1 > 0\}$ when $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\theta, 1)$; the answer there is $\Phi\big(\overline{X}\sqrt{n/(n-1)}\big)$. Now let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\theta, \sigma^2)$ with *both* unknown, so the target is

$$\psi(\theta, \sigma) = \mathbb{P}\{X_1 > 0\} = \Phi\Big(\frac{\theta}{\sigma}\Big).$$

Step 1: complete sufficient statistic. $(\overline{X},\ A)$ with $A := \sum_i (X_i - \overline{X})^2 = (n-1)S^2$, by the two-parameter exponential family criterion (Lecture 5, Ex 1).

Step 2: Rao–Blackwellise. $\mathbb{I}\{X_1 > 0\}$ is unbiased for ψ, so the UMVUE is $\mathbb{P}\big(X_1 > 0 \;\big|\; \overline{X}, A\big)$. Write

$$R := \frac{X_1 - \overline{X}}{\sqrt{A}}, \qquad \text{so} \qquad X_1 = \overline{X} + R\sqrt{A}, \qquad \{X_1 > 0\} = \Big\{R > -\frac{\overline{X}}{\sqrt{A}}\Big\}.$$

Step 3: R is ancillary, so Basu applies. Replacing X_i by $(X_i - \theta)/\sigma$ changes neither numerator nor denominator of R in a way that survives the ratio, so R has the distribution of $(Z_1 - \overline{Z})/\sqrt{\sum_j (Z_j - \overline{Z})^2}$ for $Z_j \overset{\text{iid}}{\sim} N(0,1)$ — free of (θ, σ). By Basu, $R \perp\!\!\!\perp (\overline{X}, A)$, so the conditional probability is just a distribution function of R evaluated at the observed threshold.

Step 4: the law of R. The vector $(Z_1 - \overline{Z}, \ldots, Z_n - \overline{Z})$ is spherically symmetric inside the $(n-1)$-dimensional subspace orthogonal to $\mathbf{1}$. Hence R^2 is $\frac{n-1}{n}$ times a squared cosine between a uniform random direction and a fixed one in $\mathbb{R}^{n-1}$:

$$R^2 \overset{d}{=} \frac{n-1}{n} B, \qquad B \sim \mathrm{Beta}\Big(\tfrac{1}{2}, \tfrac{n-2}{2}\Big),$$

and R is symmetric about 0 with $|R| \leq \sqrt{(n-1)/n}$. Equivalently $U := R\sqrt{n/(n-1)}$ has the symmetric density

$$f_U(u) = \frac{\Gamma\big(\frac{n-1}{2}\big)}{\sqrt{\pi}\,\Gamma\big(\frac{n-2}{2}\big)} (1 - u^2)^{\frac{n-4}{2}}, \qquad |u| < 1.$$

(Numerical check at $n = 7$: simulated $\mathbb{E}R^2 = 0.1426$ against the formula's $\frac{n-1}{n} \cdot \frac{1}{n-1} = 0.1429$, and $\max|R| = 0.919$ against $\sqrt{6/7} = 0.926$.)

Step 5: assemble. With $-\overline{X}/\sqrt{A}$ as the threshold for R, the threshold for U is $-\overline{X}\sqrt{n}/\big((n-1)S\big)$, so by symmetry of f_U,

$$\boxed{\widehat{\psi} = \mathbb{P}\big(U < W\big) = F_U(W), \qquad W := \frac{\sqrt{n}\,\overline{X}}{(n-1)S}, \qquad S = \sqrt{\tfrac{A}{n-1}},}$$

truncated in the obvious way: $\widehat{\psi} = 0$ if $W \leq -1$ and $\widehat{\psi} = 1$ if $W \geq 1$. Explicitly, for $|W| < 1$,

$$\widehat{\psi} = \frac{\Gamma\big(\frac{n-1}{2}\big)}{\sqrt{\pi}\,\Gamma\big(\frac{n-2}{2}\big)} \int_{-1}^{W} (1 - u^2)^{\frac{n-4}{2}}\, du.$$

Being unbiased and a function of the complete sufficient statistic, this is the UMVUE by Lehmann–Scheffé.

Sanity check. As $n \to \infty$, U concentrates with $\mathrm{Var}(U) \approx 1/(n-3)$, so $F_U(W) \approx \Phi\big(W\sqrt{n-3}\big) \approx \Phi(\overline{X}/S)$ — exactly the plug-in estimator of $\Phi(\theta/\sigma)$, as it must be. Note the contrast with the known-σ case: there the answer was a normal cdf; here it is a *bounded* Beta-type cdf, and the estimate is exactly 0 or 1 whenever $|\overline{X}|$ is large relative to S.

■ **Exercise 8.2.** *$X \sim \mathrm{Bin}(n,\theta)$ with prior $\pi(\theta) = \theta^{p-1}(1-\theta)^{q-1}/B(p,q)$.* How *is the Bayes estimate* $\dfrac{p+X}{p+q+n}$ *obtained?*

Answer. By Exercise 6.2 the posterior is $\mathrm{Beta}(p+x,\ q+n-x)$. The mean of a $\mathrm{Beta}(a,b)$ is $a/(a+b)$, so the Bayes estimate under squared error is

$$\mathbb{E}[\theta \mid X = x] = \frac{p+x}{(p+x)+(q+n-x)} = \frac{p+x}{p+q+n}. \qquad \checkmark$$

Written as a weighted average,

$$\frac{p+X}{p+q+n} = \frac{n}{p+q+n}\cdot\frac{X}{n} + \frac{p+q}{p+q+n}\cdot\frac{p}{p+q},$$

i.e. a convex combination of the MLE X/n and the prior mean $p/(p+q)$, with weights proportional to n and to the prior "sample size" $p+q$. This form makes the bias — and hence Exercise 6.6 — obvious.

■ **Exercise 8.3.** *Show* $\max_\theta R(\theta,T) = \dfrac{1}{4n}$ *for the UMVUE $T = X/n$.*

Answer. $T = X/n$ is unbiased, so its risk is its variance:

$$R(\theta,T) = \mathrm{Var}_\theta\left(\frac{X}{n}\right) = \frac{n\theta(1-\theta)}{n^2} = \frac{\theta(1-\theta)}{n}.$$

The parabola $\theta(1-\theta)$ on $(0,1)$ is maximised at $\theta = \frac{1}{2}$ with value $\frac{1}{4}$, so

$$\max_{0<\theta<1} R(\theta,T) = \frac{1}{4n}.$$

Compare with the minimax estimator $T^* = \dfrac{X+\sqrt{n}/2}{n+\sqrt{n}}$, whose risk is the constant $\dfrac{1}{4(1+\sqrt{n})^2}$. Since $\dfrac{1}{4(1+\sqrt{n})^2} < \dfrac{1}{4n}$ for every $n \geq 1$, **the UMVUE is not minimax** — though the gap closes as $n \to \infty$, the ratio tending to 1.

■ **Exercise 8.4.** *Complete the proof: if T_π is Bayes for π and $R(\theta,T_\pi) = c$ is constant in θ, then T_π is minimax.*

Answer. Let T be any estimator. Then

$$\sup_{\theta\in\Theta} R(\theta,T) \overset{(1)}{\geq} \int_\Theta R(\theta,T)\,\pi(\theta)\,d\theta = r(\pi,T) \overset{(2)}{\geq} r(\pi,T_\pi) \overset{(3)}{=} c \overset{(4)}{=} \sup_\theta R(\theta,T_\pi).$$

(1) A supremum dominates any average with respect to a probability measure.
(2) T_π minimises the Bayes risk, by definition of a Bayes estimate.
(3) $r(\pi,T_\pi) = \int R(\theta,T_\pi)\pi(\theta)d\theta = \int c\,\pi(\theta)d\theta = c$, because the risk is constant and π is a probability measure.
(4) The risk of T_π is the constant c.

Since T was arbitrary, $\inf_T \sup_\theta R(\theta,T) = \sup_\theta R(\theta,T_\pi)$, i.e. T_π is minimax. Such a π is called a *least favourable prior*, and an estimator with constant risk is an *equalizer*. □

■ **Exercise 8.5.** *If a generalized Bayes estimate under an improper prior is an equalizer estimate, can it be minimax?*

Answer. Sometimes yes, but not by the previous exercise's argument — and knowing exactly where that argument breaks is the point.

Where it breaks. Step (2) above needs T_π to minimise $r(\pi,\cdot)$, and steps (1) and (3) need π to be a *probability* measure. Under an improper prior, $\int \pi = \infty$: "sup $\geq$ average" is meaningless, $r(\pi, T) = \infty$ for essentially every T, and "minimising" an identically infinite quantity says nothing. So being generalized Bayes plus an equalizer is **not sufficient** on its own.

The repair. Replace the single improper prior by a sequence of *proper* priors π_k whose Bayes risks converge up to the constant risk, then use the limit-of-Bayes-rules theorem (Lecture 9): if $r(\pi_k, T_{\pi_k}) \to c$ and $\sup_\theta R(\theta, T^*) = c$, then T^* is minimax.

The example where it works. $X \sim N(\theta, 1)$, $\theta \in \mathbb{R}$. Under the improper flat prior $\pi \equiv 1$ the generalized Bayes estimate is $T^* = X$, an equalizer with constant risk 1. Taking $\pi_k = N(0, k)$ gives Bayes estimates $kX/(k+1)$ with Bayes risks $\frac{k}{k+1} \to 1$, so X is minimax. (This is Homework 2 Q1.)

The example where it fails. $X \sim \mathrm{Poi}(\lambda)$, $\lambda \in (0, \infty)$: every estimator has $\sup_\lambda R = \infty$ (Homework 2 Q2), so no finite-risk equalizer exists at all and no such conclusion is available. Similarly for $U[a, \theta]$ with θ unbounded — Exercise 9.1.

9 Lecture 9

■ **Exercise 9.1.** $X_1, \dots, X_n \overset{iid}{\sim} U[a, \theta]$ *with* $\theta \geq a$. *Show that any estimate of* θ *has maximum risk* ∞.

Answer. Use the conjugate Pareto priors from the same lecture and let the scale run off to infinity.

Setup. $X_{(n)}$ is sufficient with density nx^{n-1}/θ^n on $[a, \theta]$. Take the Pareto prior with scale m ($\geq a$) and shape $\alpha > 3$,

$$\pi_m(\theta) = \frac{\alpha m^\alpha}{\theta^{\alpha+1}}, \qquad \theta > m.$$

Posterior. $\pi(\theta \mid \mathbf{x}) \propto \theta^{-n}\mathbb{I}[\theta > x_{(n)}] \cdot \theta^{-\alpha-1}\mathbb{I}[\theta > m] = \theta^{-(\alpha+n)-1}$ for $\theta > M := \max(m, x_{(n)})$: again Pareto, with shape $\alpha + n$ and scale M.

Bayes risk. A Pareto(β, M) variable has variance $\dfrac{\beta M^2}{(\beta-1)^2(\beta-2)}$ for $\beta > 2$, so with $\beta = \alpha + n$,

$$r(\pi_m) = \mathbb{E}_m\big[\,\mathrm{Var}(\theta \mid X)\big] = \frac{(\alpha+n)}{(\alpha+n-1)^2(\alpha+n-2)}\,\mathbb{E}\big[M^2\big] \;\geq\; \frac{(\alpha+n)\,m^2}{(\alpha+n-1)^2(\alpha+n-2)},$$

using $M \geq m$. With α, n fixed, the prefactor is a positive constant, so

$$r(\pi_m) \;\geq\; C(\alpha, n)\, m^2 \;\xrightarrow[m\to\infty]{}\; \infty.$$

Conclusion. For any estimator T and every m,

$$\sup_{\theta \geq a} R(\theta, T) \;\geq\; r(\pi_m, T) \;\geq\; r(\pi_m) \;\geq\; Cm^2,$$

so letting $m \to \infty$ gives $\sup_\theta R(\theta, T) = \infty$. **Every estimator has infinite maximum risk**; the minimax value of the problem is $+\infty$ and no minimax estimator exists in any useful sense. □

The mechanism is the same as in the Poisson problem: an *unbounded* parameter space in which the achievable variance grows without bound. Compare $\mathrm{Bin}(n, \theta)$, where $\theta \in (0, 1)$ is bounded and a genuine minimax estimator exists.

10 Lecture 10

■ **Exercise 10.1.** *Work out the consistency of $\hat{\alpha}_{LS}$ in $Y_i = \alpha + \beta x_i + \varepsilon_i$. Assume $\bar{x}$ converges to a constant.*

Answer. Write $S_{xx} = \sum_i (x_i - \bar{x})^2$. Then $\hat{\alpha} = \overline{Y} - \hat{\beta}\bar{x}$ with $\hat{\beta} = \sum_i (x_i - \bar{x}) Y_i / S_{xx}$.

Unbiasedness. $\mathbb{E}\hat{\alpha} = \mathbb{E}\overline{Y} - \bar{x}\,\mathbb{E}\hat{\beta} = (\alpha + \beta\bar{x}) - \bar{x}\beta = \alpha$.

Variance. First,

$$\operatorname{Cov}(\overline{Y}, \hat{\beta}) = \frac{1}{n\,S_{xx}} \sum_i (x_i - \bar{x}) \operatorname{Var}(Y_i) = \frac{\sigma^2}{n\,S_{xx}} \sum_i (x_i - \bar{x}) = 0,$$

because $\sum_i (x_i - \bar{x}) = 0$. Hence the cross term drops and

$$\boxed{\operatorname{Var}(\hat{\alpha}) = \operatorname{Var}(\overline{Y}) + \bar{x}^2 \operatorname{Var}(\hat{\beta}) = \frac{\sigma^2}{n} + \frac{\bar{x}^2\sigma^2}{S_{xx}}}$$

Consistency. Suppose $\bar{x} \to c$ finite and $S_{xx} \to \infty$ (the condition that already gives $\hat{\beta}$'s consistency). Then

$$\operatorname{Var}(\hat{\alpha}) = \frac{\sigma^2}{n} + \frac{\bar{x}^2\sigma^2}{S_{xx}} \longrightarrow 0,$$

since both terms vanish. With zero bias, Chebyshev gives $\hat{\alpha} \xrightarrow{p} \alpha$.

Both hypotheses are needed. If $\bar{x} \to \infty$ fast enough that $\bar{x}^2/S_{xx} \not\to 0$, the intercept is not consistently estimable even though the slope is: the data live far from $x = 0$, so extrapolating back to the intercept never stabilises. This is the familiar warning that an intercept is only well determined when the design is spread *around* the origin.

■ **Exercise 10.2.** *Show the consistency of $\hat{\theta}_{MLE}$.*

Answer. The notes give the answer in words — "yes, if regularity conditions on $f(x, \theta)$ hold and the ML equation has a unique solution for each n" — and the standard argument (Wald's) is the following.

Assumptions. (i) *Identifiability*: $\theta \neq \theta_0 \Rightarrow f(\cdot \mid \theta) \neq f(\cdot \mid \theta_0)$; (ii) common support, free of θ; (iii) Θ compact (or the maximiser eventually lies in a compact set); (iv) $\theta \mapsto \log f(x \mid \theta)$ continuous, with an integrable envelope so that a uniform law of large numbers applies.

Step 1: the population criterion is maximised at the truth. Let θ_0 be the true value and define $M(\theta) = \mathbb{E}_{\theta_0} \log f(X \mid \theta)$. By Jensen's inequality applied to the concave log,

$$M(\theta) - M(\theta_0) = \mathbb{E}_{\theta_0} \log \frac{f(X \mid \theta)}{f(X \mid \theta_0)} \;\leq\; \log \mathbb{E}_{\theta_0} \frac{f(X \mid \theta)}{f(X \mid \theta_0)} = \log \int f(x \mid \theta)\, dx = \log 1 = 0,$$

with equality iff $f(X \mid \theta)/f(X \mid \theta_0)$ is a.s. constant, i.e. iff the two densities coincide, i.e. iff $\theta = \theta_0$ by identifiability. So M has a *strict* maximum at θ_0. (The quantity $-[M(\theta) - M(\theta_0)]$ is the Kullback–Leibler divergence.)

Step 2: the sample criterion converges uniformly. With $M_n(\theta) = \frac{1}{n}\sum_{i=1}^n \log f(X_i \mid \theta)$, the law of large numbers gives $M_n(\theta) \to M(\theta)$ for each θ, and under (iii)–(iv) the convergence is uniform: $\sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| \xrightarrow{p} 0$.

Step 3: the argmax converges. Fix $\varepsilon > 0$ and let $\delta = M(\theta_0) - \sup_{|\theta - \theta_0| \geq \varepsilon} M(\theta) > 0$ by Step 1 and compactness. On the event $\sup_\theta |M_n - M| < \delta/2$, which has probability $\to 1$,

$$M_n(\hat{\theta}_n) \geq M_n(\theta_0) > M(\theta_0) - \tfrac{\delta}{2} \quad \text{while} \quad M_n(\theta) < M(\theta) + \tfrac{\delta}{2} \leq M(\theta_0) - \tfrac{\delta}{2} \;\text{ for } |\theta - \theta_0| \geq \varepsilon,$$

so $\hat{\theta}_n$ cannot lie outside the ε-ball. Hence $\mathbb{P}(|\hat{\theta}_n - \theta_0| < \varepsilon) \to 1$, i.e. $\hat{\theta}_n \xrightarrow{p} \theta_0$. □

The two examples from the lecture.

- $U(0,\theta)$: the support depends on θ, so assumption (ii) fails and the theorem does not apply — yet $\hat\theta_n = X_{(n)}$ *is* consistent, shown directly from the cdf: $\mathbb{P}(\hat\theta_n < \theta - \varepsilon) = (1-\varepsilon/\theta)^n \to 0$ and $\mathbb{P}(\hat\theta_n > \theta + \varepsilon) = 0$. It is even n-consistent, not merely $\sqrt{n}$-consistent. Regularity is sufficient, not necessary.
- Cauchy: all the assumptions hold, so the MLE is consistent; but the likelihood equation $\sum_i \frac{2(x_i-\theta)}{1+(x_i-\theta)^2} = 0$ can have several roots, so "the MLE" means the *global* maximiser. Starting Newton–Raphson from the sample median (itself consistent, by the argument worked in this lecture) is what makes the numerical problem well behaved — see Homework 2 Q4.

Companions: `PI_Exam_Revision.pdf` *(theory, proofs, cheat sheets) and* `PI_Assignment_Solutions.pdf` *(Homeworks 1–3).*